

Minimal metrics and interior injectivity for nef adjoints

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Abstract

Let (H, Θ) be a projective complex klt pair with effective rational boundary and nef $\mathbb{Q}$ -Cartier adjoint. On any projective log resolution, minimal semipositive metrics on the pulled-back adjoint exist and have zero Lelong numbers everywhere. On smooth projective complex varieties, we also prove an H^1 -injectivity theorem for rational interior boundaries with simple-normal-crossing support when both endpoint bundles carry zero-Lelong semipositive metrics.

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1 Introduction

A nef line bundle admits smooth metrics whose curvatures approach semipositivity. A limit of those metrics may acquire singularities. For an adjoint bundle, the question is whether the canonical and boundary terms prevent logarithmic poles in a least singular semipositive metric. We prove that they do for projective klt pairs, and use zero-Lelong endpoint metrics to establish an injectivity theorem in ordinary coherent cohomology.

We use the convention that a local metric weight ϕ gives squared norm $|\xi|^2 e^{-\phi}$. Semipositivity means that its curvature current $i\partial\bar{\partial}\phi$ is nonnegative. Such a metric has *minimal singularities* if every other semipositive weight ψ on the same line bundle satisfies $\psi \leq \phi + C_\psi$ for a constant C_ψ . Weights on rational line bundles are defined by clearing denominators and dividing by the same positive integer. Lelong numbers measure the logarithmic singularities of these weights; in particular, zero Lelong numbers imply local integrability of $e^{-t\phi}$ for every fixed $t > 0$.

For a normal variety H and an effective rational divisor Θ , the klt condition means that $K_H + \Theta$ is $\mathbb{Q}$ -Cartier and, on a log resolution $\pi : V \rightarrow H$, the crossing divisor B in $\pi^*(K_H + \Theta) = K_V + B$ has every coefficient less than one. The actual rational line bundle, rather than its numerical class, is the object in the following theorem.

Theorem 1.1. *Let (H, Θ) be a normal connected projective complex klt pair, with Θ an effective rational divisor. Suppose that $D_H = K_H + \Theta$ is a nef $\mathbb{Q}$ -Cartier divisor. For every*

projective log resolution $\pi : V \rightarrow H$, the rational line bundle $N = \pi^* D_H$ has the following property: every semipositive singular Hermitian metric with minimal singularities on N has Lelong number zero at every point of V . Such a metric exists. Neither K_H nor Θ is required to be separately $\mathbb{Q}$ -Cartier.

Consequently, for every minimal weight ϕ on $\pi^*(K_H + \Theta)$,

$$\mathcal{I}(t\phi) = \mathcal{O}_V \quad (t > 0),$$

where $\mathcal{I}(t\phi)$ consists of holomorphic germs f for which $|f|^2 e^{-t\phi}$ is locally integrable. We record the precise independence of the metric and the Cartier multiple in Corollary 2.4.

Demailly–Peternell–Schneider constructed minimal-singularity metrics on pseudoeffective line bundles and proved zero Lelong numbers in the big and nef case [2, Theorem 1.5 and Proposition 1.7]. Nefness alone does not give this conclusion. For example, on a ruled surface over an elliptic curve, a nef divisor bundle can have its divisor metric as a minimal metric, with a positive Lelong number along the divisor [14, Corollary 1.2 and Example 3.5]. Gongyo–Matsumura asked how to construct a zero-Lelong semipositive metric on a nef log canonical bundle in their study of injectivity and abundance [11, Question 5.6]. Theorem 1.1 gives a positive answer in the projective klt setting on every projective log resolution, without bigness or nonvanishing. Once one zero-Lelong semipositive metric exists, every minimal metric is no more singular and therefore also has zero Lelong numbers. Thus the universal assertion about minimal metrics expresses the same existence conclusion on each pullback bundle.

1.1 Interior boundaries and restriction of sections

Our second result concerns a family of effective rational divisors between two fixed endpoints. Their coefficients may cross integers. The resulting changes in the integral part of the divisor give natural inclusions of line bundles. Zero-Lelong metrics at the endpoints control the induced map on ordinary H^1 .

Theorem 1.2. *Let V be a smooth projective complex variety, let L be an integral divisor, and let $0 \leq C_0 \leq C_2$ be effective rational divisors whose combined support is simple normal crossing. Suppose that both rational line bundles $L - C_0$ and $L - C_2$ admit singular Hermitian metrics of semipositive curvature with zero Lelong numbers at every point. For a rational number $0 < \lambda < 1$, set*

$$C_1 = (1 - \lambda)C_0 + \lambda C_2, \quad L_i = L - \lfloor C_i \rfloor \quad (i = 0, 1).$$

Then the natural inclusion of line bundles induces an injection

$$H^1(V, \mathcal{O}_V(K_V + L_1)) \longrightarrow H^1(V, \mathcal{O}_V(K_V + L_0)).$$

The strict inequality $\lambda < 1$ provides a positive coefficient in the curvature comparison. Effectivity and the common crossing support provide integrability for the fractional boundary weights. The theorem concerns the displayed integral line bundles and the natural sheaf map; its endpoint assumptions are on the actual rational bundles $L - C_i$.

One direct consequence explains the connection with restriction of sections. Put $E = \lfloor C_1 \rfloor - \lfloor C_0 \rfloor$, an effective integral divisor, which may be nonreduced. If $E \neq 0$, then

$$H^0(V, \mathcal{O}_V(K_V + L_0)) \longrightarrow H^0(E, \mathcal{O}_V(K_V + L_0)|_E) \quad \text{is surjective.} \quad (1.1)$$

Indeed, the connecting homomorphism for $0 \rightarrow \mathcal{O}_V(K_V + L_1) \rightarrow \mathcal{O}_V(K_V + L_0) \rightarrow \mathcal{O}_V(K_V + L_0)|_E \rightarrow 0$ has image equal to the kernel of the injective H^1 map. Thus the comparison lifts sections from the entire divisor scheme E at once.

Classical root-cover injectivity, including a reduced boundary, appears in Esnault–Viehweg [7, Theorem 5.1]. Fujino’s transcendental approach develops complete metrics on an analytic complement and comparison with coherent cohomology [10, Section 3, Lemmas 3.1–3.2 and Claim 1]. Matsumura extends analytic injectivity to metrics with transcendental singularities and obtains uniform estimates for primitives [19, Theorem 1.3 and Sections 5.2–5.3]. Our proof uses the same harmonic-form and local-primitive methods, with a comparison of two endpoint weights. We prove the required uniform primitive estimate and the return to ordinary cohomology explicitly.

1.2 The two analytic arguments

For the metric theorem, write $N = \pi^*(K_H + \Theta) = K_V + B$. The adjoint hypothesis enters twice. The klt coefficients make the adjoint density integrable, even where B has negative exceptional coefficients. Klt vanishing and nefness then give one ample bundle A_H on H whose pullback $A = \pi^*A_H$ makes $mN + A$ globally generated for every positive integer m with $m(K_H + \Theta)$ Cartier. Normalize a section s by its integral

$$Q_m(s) = \int_V |s|^{2/m} e^{-a/m-b},$$

where a is a smooth weight of A and b the divisor weight of B . Choose a normalized section maximizing its value at a proposed positive-Lelong point.

The $L^{2/m}$ normalization has its origin in Narasimhan–Simha’s work on ample canonical bundles [20]; its role in the later envelope construction is explained in [1, Section 5]. Tsuji constructs canonical singular metrics using normalized sections of $A + mK_X$ with a fixed sufficiently ample twist [22, Sections 1.2 and 2]. Berman–Demailly develop integral-normalized adjoint envelopes and their algebraic approximation [1, Definition 5.3 and Proposition 5.19]. Their general pseudoeffective approximation allows twists $p_m A$ with $p_m \rightarrow \infty$ and $p_m/m \rightarrow 0$ [1, Equation (5.24)]. Here the fixed twist and the klt density support an extremal argument: a localization estimate produces a better normalized section whenever the minimal weight has a positive Lelong number.

The normalized extremal sections have a semipositive limit after the twist is divided by m . Minimality bounds that limit by the chosen minimal weight. Consequently a fixed sublevel set of the difference of the weights has adjoint mass tending to zero. A localized $\bar{\partial}$ equation, solved on an affine open set, produces a holomorphic competitor whose weighted norm is bounded by that mass. The exact localization constant is independent of m and of the affine open set. Positive Lelong number forces this competitor to have the same value at the chosen point. A Hölder estimate whose final bound is independent of m makes its normalization strictly smaller, contradicting extremality.

Two details keep the construction uniform and valid on a log resolution. The weak limits in the localization lemma are taken in fixed weighted Hilbert spaces. Extension across negative exceptional coefficients is performed on the normal base H in the actual Cartier bundle $m(K_H + \Theta) + A_H$, and then pulled back. The complete Kähler L^2 estimate supplies the local analytic solution [4, Theorem 5.1]; the localization and extension arguments are given in Section 2.

For injectivity, regularize both endpoint metrics, retaining arbitrarily small curvature loss and uniform integrability at every fixed exponent. A logarithm of a sum of endpoint weights makes the inclusion of line bundles a contraction. On the complement of the crossing divisor, a complete Kähler metric with bounded local potentials allows local L^2 solutions. Their holomorphic differences extend across the divisor and define ordinary Čech classes. An open-mapping argument on one fixed finite cover gives a uniform bound for primitives of forms whose ordinary class vanishes.

A Bochner comparison then makes the image of a harmonic representative small. All approximating representatives embed into a single fixed Hilbert space; weak convergence there preserves the original ordinary cohomology class. This is the final step from weighted estimates on the complement to the coherent-cohomology injection. The argument is independent of the adjoint metric theorem: it uses only the endpoint metrics stated in Theorem 1.2.

1.3 A fourfold application and organization

The metric theorem also gives a short abundance argument for a projective complex klt fourfold (X, Δ) with nef rational adjoint and $\chi(X, \mathcal{O}_X) \neq 0$. Lazić–Petrernell’s nonvanishing theorem uses generalized algebraic singularities of a semipositive metric [17, Corollary D]; zero Lelong numbers satisfy that condition with no divisorial part. With a first section obtained, Gongyo–Matsumura’s criterion gives semiampleness [11, Corollary 5.3]. The precise application is Corollary 4.1; it has no numerical-dimension restriction. For nef adjoints of numerical dimension at most one, Liu–Xu already obtain good minimal models for projective klt pairs of dimension at most four with nonzero Euler characteristic [18, Corollary 5.2]; in the nef case this gives semiampleness. Lazić’s work supplies a semiampleness criterion using uniform multiplier-ideal comparisons, and an algebraic approximation theorem for supercanonical currents, under their respective hypotheses [16, Theorems A–B].

In the compact Kähler setting, Höring–Lazić–Lehn obtain nonvanishing for non-uniruled $\mathbb{Q}$ -factorial klt pairs with nef adjoint of numerical dimension one and nonzero Euler characteristic [13, Theorem A]. Under these hypotheses in dimension four, they obtain semiampleness when the minimal pullback current has zero Lelong numbers [13, Corollary B].

Section 2 proves the metric theorem, the arbitrary-data localization estimate and the multiplier-ideal consequence. Section 3 proves the independent interior injectivity theorem. Section 4 states the two published fourfold criteria and applies them to the same minimal metric.

2 Zero Lelong numbers for nef klt adjoints

We prove Theorem 1.1. The construction uses sections only after one fixed twist pulled back from an ample bundle on the normal base. Its central estimate must reduce a normalized section’s mass without changing its value at a point of positive Lelong number. We first recall the top-degree L^2 estimate, then construct the normalized extremal sections. The affine localization lemma turns their small sublevel mass into the required competitor.

2.1 Weights and the complete L^2 estimate

A local weight φ of a Hermitian metric gives the squared norm $|\xi|^2 e^{-\varphi}$ in its local frame. Semipositive curvature means that the local weights are plurisubharmonic. A metric on a rational line bundle is defined by taking a positive integral tensor power and dividing its weights by that integer. All identifications below are identifications of rational line bundles, rather than of numerical classes.

If a weight η is a weight on K_V , the expression e^η denotes a density: in a canonical coordinate frame it is multiplied by the corresponding coordinate volume. Likewise, for an F -valued top form v , the expression $|v|^2 e^{-\psi}$, with ψ a weight on F , denotes a density. These densities are invariant under changes of coordinates and frames. In particular, the integral of the latter does not depend on an auxiliary Kähler metric.

We shall use the following form of the complete L^2 theorem. The matrix observation in its statement will be responsible for the uniform constants.

Lemma 2.1. *Let (M, ω_M) be a complete Kähler manifold of dimension n , and let F be a holomorphic line bundle with a smooth Hermitian metric of local weight ψ and strictly positive curvature θ . Write Λ_{ω_M} for the adjoint of wedging with ω_M . If β is an F -valued, $\bar{\partial}$ -closed $(n, 1)$ form which is square integrable and satisfies*

$$I(\beta) = \int_M \langle [\theta, \Lambda_{\omega_M}]^{-1} \beta, \beta \rangle_{\psi, \omega_M} dV_{\omega_M} < \infty,$$

then there is an F -valued $(n, 0)$ form v with

$$\bar{\partial}v = \beta, \quad \int_M |v|^2 e^{-\psi} \leq I(\beta).$$

In local coordinates, the inverse-curvature density in $I(\beta)$ is the inverse Hermitian curvature matrix acting on the remaining $(0, 1)$ covector, multiplied by the coordinate top-form density. It is independent of the complete background metric. Consequently:

- (i) *if $\theta \geq \omega_0 > 0$, this density is bounded by the $(n, 1)$ norm density computed with ω_0 ;*
- (ii) *if g is a smooth real function and $\theta \geq a i \partial \bar{\partial} g \wedge \bar{\partial} g$ with $a > 0$, the inverse-curvature squared norm of $\bar{\partial} g$ is at most a^{-1} .*

Proof. The existence statement is the complete Kähler L^2 theorem with positive curvature operator and finite inverse-curvature integral [4, Theorem 5.1, p. 33]. It requires pointwise positivity and the displayed integral, not a uniform lower spectral bound relative to ω_M .

For the density calculation, write a top-form-valued $(0, 1)$ form as $\sum_j \beta_j dz_1 \wedge \cdots \wedge dz_n \wedge d\bar{z}_j$. The determinant in the squared norm of the top form cancels the volume determinant. Diagonalizing the curvature relative to ω_M shows that applying the inverse commutator leaves $\sum_{j,k} (\theta^{-1})^{jk} \beta_j \bar{\beta}_k$ against the coordinate volume and line weight. All formulas use the same standard normalization of forms and volume. Matrix order gives $\theta^{-1} \leq \omega_0^{-1}$ in (i). For (ii), conjugate the rank-one inequality $a \partial g \otimes \bar{\partial} g \leq \theta$ by $\theta^{-1/2}$. $\square$

2.2 The fixed twist and normalization

The zero-dimensional case is immediate, so put $n = \dim V > 0$ and fix a Kähler form ω on V . Choose compatible canonical divisors and write

$$N = K_V + B. \tag{2.1}$$

The support of B is simple normal crossing, every coefficient is strictly less than one, and the negative part B^- is exceptional over H . The last assertion follows because the nonexceptional coefficients are those of the effective boundary Θ .

Let $b = [B]$ be the divisor weight, a rational linear combination of logarithms of squared absolute values of local divisor equations. In crossing coordinates it has the form

$$b = \sum_{j=1}^a \beta_j \log |z_j|^2 + O(1), \quad \beta_j < 1.$$

Thus e^{-b} is locally integrable. Negative coefficients cause no failure of this integrability statement.

Fix a very ample Cartier divisor H_0 on H , and set $A_H = (n+1)H_0$ and $A = \pi^*A_H$. Give A a smooth semipositive metric with weight a . For every positive integer m for which mD_H is Cartier, the line bundle $mD_H + A_H$ is globally generated. Indeed, for $1 \leq j \leq n$,

$$mD_H + A_H - jH_0 - (K_H + \Theta) = (m-1)D_H + (n+1-j)H_0$$

is ample. Klt Kawamata–Viehweg vanishing, in the form of [9, Theorem 3.2], gives the cohomology vanishings for Castelnuovo–Mumford regularity. The global-generation conclusion of [15, Theorem 1.8.5(i)] then applies. Pullback proves that $mN + A$ is globally generated as well.

Since N is nef, it is pseudo-effective. Choose a semipositive metric ϕ on N with minimal singularities. Such a metric is obtained as the upper envelope, relative to a smooth reference weight, of the normalized plurisubharmonic weights; see [2, Definition 1.4 and Theorem 1.5]. Its defining property is that every other semipositive weight ψ on N satisfies

$$\psi \leq \phi + C_\psi \quad (2.2)$$

for a constant C_ψ . We shall show that ϕ has zero Lelong numbers.

Suppose, to the contrary, that its Lelong number at $p \in V$ is positive. For each allowed m define

$$Q_m(s) = \int_V |s|^{2/m} e^{-a/m-b}, \quad s \in H^0(V, mN + A). \quad (2.3)$$

The integrand is a canonical density by Equation (2.1). The integral is finite, continuous in s , positive for $s \neq 0$, and homogeneous of degree $2/m$. These facts also show that $Q_m(s) = 1$ is a compact subset of the finite-dimensional section space: restrict Q_m first to the unit sphere of any vector-space norm and use its positive minimum. Choose s_m maximizing the absolute evaluation at p on this compact set, using any fixed norm on the one-dimensional fiber for this m . Global generation implies

$$Q_m(s_m) = 1, \quad s_m(p) \neq 0. \quad (2.4)$$

Put $\tau_m = m^{-1} \log |s_m|^2$, a weight on $N + A/m$.

Claim 2.2. *After passage to an unbounded subsequence of allowed integers, the weights $\tau_m - a/m$ are uniformly locally bounded above and converge locally in L^1 and almost everywhere to a semipositive weight τ_∞ on N . This compactness assertion holds for every sequence of sections with $Q_m(s_m) = 1$; it does not require their extremality at p .*

Proof. Choose a smooth reference weight ϕ_{ref} on N and write $u_m = \tau_m - a/m - \phi_{\text{ref}}$. These are global quasi-psh functions whose complex Hessians have one common lower bound $-C\omega$ on the compact connected manifold V . Normalize them by subtracting $M_m = \sup_V u_m$. The compactness theorem for sup-normalized quasi-psh functions [12, Proposition 2.7] gives, along a subsequence, local L^1 and almost-everywhere convergence of $u_m - M_m$ to a quasi-psh function u_∞ that is finite almost everywhere. In particular the normalized functions do not converge identically to $-\infty$.

The fixed density $e^{\phi_{\text{ref}}-b}$ is integrable. Since $u_m - M_m \leq 0$, dominated convergence gives

$$J_m := \int_V e^{u_m - M_m} e^{\phi_{\text{ref}} - b} \longrightarrow J_\infty := \int_V e^{u_\infty} e^{\phi_{\text{ref}} - b} \in (0, \infty).$$

The positivity uses finiteness of u_∞ almost everywhere, not a pointwise lower bound. The normalization in Equation (2.4) says $e^{M_m} J_m = 1$. Thus M_m converges to a finite real number, proving the asserted upper bounds and convergence without the sup normalization. Finally, $i\partial\bar{\partial}(\tau_m - a/m) \geq -m^{-1}i\partial\bar{\partial}a$; the error tends to zero. The limit weight is therefore semipositive. $\square$

Define the globally scalar difference, almost everywhere,

$$g_m = \phi + a/m - \tau_m.$$

Its values on the zero divisor of s_m may be assigned arbitrarily when taking integrals; that divisor has measure zero. By Equation (2.2), $\tau_\infty \leq \phi + C_\infty$. Choose once and for all $R > C_\infty + 1$. The preceding convergence and domination imply

$$\varepsilon_m := \int_{\{g_m < -R\}} e^{\tau_m - a/m - b} \longrightarrow 0. \quad (2.5)$$

Indeed, the indicators tend to zero almost everywhere, because $\phi - \tau_\infty \geq -C_\infty$ where both weights are finite; the densities have one integrable majorant on a fixed finite coordinate cover.

It is now enough to construct sections $w_m \in H^0(V, mN + A)$ such that

$$w_m(p) = s_m(p), \quad Q_m(w_m)^m \leq C_0 \varepsilon_m,$$

where C_0 is independent of m . For large m this gives $Q_m(w_m) < 1$, so scalar normalization would produce a section of gauge one with larger absolute evaluation at p . We next construct these competitors and prove the two displayed properties.

2.3 A singular weighted equation with a fixed constant

We will cut off s_m on $\{g_m < -R\}$ and correct the resulting $\bar{\partial}$ error. The weight for this equation must serve two purposes. Its singularity near p must force the correction to vanish at p after the holomorphic competitor has been extended. Its growth where g_m is positive must also allow a uniform conversion from the quadratic estimate to Q_m . The first requirement determines its slope c near $-\infty$; a fixed upper slope bound d will give the second.

Positive Lelong number gives local coordinates at p and a sufficiently small fixed $\alpha > 0$ with $\phi(z) \leq \alpha \log |z|^2 + O(1)$. On a cone of positive angular measure on which all crossing coordinates have modulus comparable to $|z|$, the weight $e^{-c\phi - b}$ is bounded below by a positive constant times $|z|^{-2(c\alpha + \sum \beta_j)}$. Choose a fixed $c > 0$ large enough that $c\alpha + \sum \beta_j \geq n$. Then

$$e^{-c\phi - b} \quad \text{is not locally integrable at } p. \quad (2.6)$$

This choice is valid even if some β_j are negative.

Choose a smooth function $0 \leq \chi \leq 1$ equal to one on $(-\infty, -R - 1]$ and to zero on $[-R, \infty)$. Choose a smooth convex increasing function f which is affine of slope c near $-\infty$, has $f'' > 0$ on the closed transition interval $[-R - 1, -R]$, and satisfies $c \leq f' \leq d$ for one fixed d . Such a function is obtained by prescribing a nonnegative compactly supported second derivative positive on that interval. In particular there is a fixed constant C_f with

$$f(t) \leq d \max(0, t) + C_f \quad (t \in \mathbb{R}). \quad (2.7)$$

All of R, c, χ, f, d, C_f remain fixed as m varies.

For a fixed m , choose a dense smooth affine open $U = U_m \subset V$ avoiding the supports of B and $\text{div}(s_m)$. This open set need not contain p . We will recover the value at p only after extending the resulting holomorphic section to all of V . On U the function $g = g_m$ is psh, since the weight τ_m is locally pluriharmonic there and the metrics ϕ, a are semipositive. Put

$$F_m = mN + A - K_V, \quad S = (m - 1)\tau_m + a/m + b. \quad (2.8)$$

This is a smooth semipositive weight on $F_m|_U$. Divisor weights and τ_m are pluriharmonic on U , so its curvature is $m^{-1}i\partial\bar{\partial}a$. We regard s_m as an F_m -valued top form.

Lemma 2.3 (Localization of top forms). *Let U be a smooth connected affine complex variety of dimension $n \geq 1$. Let F be a holomorphic line bundle on U with a smooth semipositive metric of weight S , let g be a global plurisubharmonic function not identically $-\infty$, and let s be a holomorphic F -valued $(n, 0)$ -form. Fix $R \in \mathbb{R}$, a smooth cutoff $0 \leq \chi \leq 1$ equal to one for $t \leq -R - 1$ and zero for $t \geq -R$, and a smooth convex function f that is affine of positive slope c near $-\infty$, with $c \leq f' \leq d$ for some d and $f'' > 0$ on $[-R - 1, -R]$. If*

$$M = \int_{\{g < -R\}} |s|^2 e^{-S} < \infty,$$

there is a measurable F -valued top form u such that

$$\bar{\partial}u = \bar{\partial}(\chi(g)s), \quad \int_U |u|^2 e^{-S-f(g)} \leq C_{\chi,f} M, \quad (2.9)$$

where the equation is distributional and

$$C_{\chi,f} = \max_{[-R-1, -R]} \frac{|\chi'|^2 e^{-f}}{f''}. \quad (2.10)$$

The constant is independent of U, F, S, g, s and of the auxiliary complete Kähler metrics. At $g = -\infty$ we set $\chi(g) = 1$ and $f(g) = -\infty$; that locus has measure zero.

Proof. Embed U as a closed complex submanifold of some $\mathbb{C}^a$ and let $\rho = |z|^2|_U$. The tubular-neighborhood theorem supplies a holomorphic retraction of a neighborhood onto U [6]; see also the formulation in [8, Theorem 3.1]. Compose g with the retraction. Radial convolution in that neighborhood, at radii decreasing to zero, gives smooth psh functions $g_j \downarrow g$ on common smaller domains. More precisely, choose increasing regular exhaustion values $a_j \rightarrow \infty$ and $\Omega_j = \{\rho < a_j\}$. Use one fixed nonnegative smooth radial averaging kernel of total mass one. The convolution radius at stage j is at most $1/j$, is small enough to work near $\overline{\Omega_j}$, and is smaller than all previous radii. Radial averages of a psh function are monotone in the radius, so these choices give monotonicity wherever two of the functions are defined. They also give $g_j \geq g$.

Each Ω_j has a complete Kähler metric dominating $i\partial\bar{\partial}\rho$, for example the metric with potential $\rho - \log(a_j - \rho)$. For a decreasing positive sequence $\delta_j \rightarrow 0$, use the line weight

$$P_j = S + f(g_j) + \delta_j \rho.$$

It is smooth near $\overline{\Omega_j}$, its curvature is strictly positive, and

$$i\partial\bar{\partial}P_j \geq f''(g_j)i\partial g_j \wedge \bar{\partial}g_j.$$

The smooth closed datum $\beta_j = \chi'(g_j)\bar{\partial}g_j \wedge s$ is square integrable for the complete metric. To see this even near the boundary of Ω_j , use the top-degree density calculation: enlarging the background metric decreases the $(n, 1)$ density, and all coefficients and line weights are smooth on a neighborhood of the relatively compact closure.

By Lemma 2.1, the inverse-curvature integral is at most

$$C \int_{\Omega_j \cap \{g_j < -R\}} |s|^2 e^{-S} \leq C \int_{\{g < -R\}} |s|^2 e^{-S}, \quad C = \sup_{[-R-1, -R]} \frac{|\chi'|^2 e^{-f}}{f''}.$$

The nonnegative term $\delta_j \rho$ only decreases the integrand. The last integral is M by definition. Thus we obtain solutions u_j on Ω_j with $\int_{\Omega_j} |u_j|^2 e^{-P_j} \leq CM$. The background complete metrics have introduced no comparison factor.

For completeness, the singular passage uses fixed Hilbert spaces. If $k \leq l$ are fixed and $j \geq l$, then on Ω_k we have $P_j \leq P_l$, and hence

$$\int_{\Omega_k} |u_j|^2 e^{-P_l} \leq CM.$$

Take a diagonal weakly convergent subsequence in these countably many fixed earlier smooth weighted spaces. Their limits agree as distributions on overlaps, and define u on U . Weak lower semicontinuity, then monotone convergence as $l \rightarrow \infty$, gives

$$\int_{\Omega_k} |u|^2 e^{-S-f(g)} \leq CM.$$

Exhausting U proves the required global inequality. Finally $\chi(g_j)s \rightarrow \chi(g)s$ locally in L^2 by bounded convergence, where $\chi(-\infty) = 1$. Thus the equations pass to distributions and give Equation (2.9). $\square$

We apply the lemma to the adjoint data already fixed on U . The exact density identity is

$$|s_m|^2 e^{-S} = e^{\tau_m - a/m - b}, \quad (2.11)$$

so its sublevel mass is $M = \varepsilon_m$. Applying Lemma 2.3 with $F = F_m$, $g = g_m$, and $s = s_m$ gives an F_m -valued top form u on U with

$$\bar{\partial}u = \bar{\partial}(\chi(g)s_m), \quad \int_U |u|^2 e^{-S-f(g)} \leq C\varepsilon_m, \quad (2.12)$$

where $C = C_{\chi,f}$ is independent of m and of U .

2.4 Extension, evaluation, and the extremal contradiction

For $m > d + 1$ set

$$G = S + d \max(0, g), \quad w_m = \chi(g)s_m - u \quad \text{on } U.$$

Equation (2.12) shows that w_m is holomorphic. Since $G \geq S$, the cutoff term has G -norm squared at most ε_m . Since $S + f(g) \leq G + C_f$, the G -norm squared of u is at most $e^{C_f} C \varepsilon_m$. Therefore

$$\int_U |w_m|^2 e^{-G} \leq C' \varepsilon_m, \quad C' = 2 + 2e^{C_f} C. \quad (2.13)$$

This constant is independent of m .

To extend the section, rewrite its weight as

$$G = b + a/m + (m - 1 - d)\tau_m + d \max\{\tau_m, \phi + a/m\}. \quad (2.14)$$

It is locally bounded above away from $\text{Supp } B^-$: the coefficient $m - 1 - d$ is positive, the psh weights in the last two terms are locally bounded above, and b has that property away from its negative components. Consequently Equation (2.13) gives ordinary local L^2 bounds there. Holomorphic L^2 removal across analytic sets extends w_m to $V \setminus \text{Supp } B^-$.

There is an open $H^\circ \subset H$, with complement of codimension at least two, over which π is an isomorphism. It misses the image of B^- . The section just obtained descends on H° to a section of the actual Cartier bundle $\mathcal{O}_H(mD_H + A_H)$. Normality extends it across $H \setminus H^\circ$: after trivializing that line bundle, this is the Hartogs extension of regular holomorphic functions on a normal space. Pullback now gives a global section, again denoted w_m , of $mN + A$. By projectivity and GAGA it is an algebraic global section as well. It agrees

with the constructed section on U by the identity theorem. In particular Equation (2.13) continues to concern the same section. No estimate across the negative exceptional divisor was needed for this extension.

Near p , the weight τ_m is smooth by Equation (2.4), whereas $g \rightarrow -\infty$. Thus $\chi(g) = 1$ and $f(g) = cg + \text{constant}$ near p on U , so

$$u = s_m - w_m, \quad S + f(g) = b + c\phi + O_m(1).$$

If $s_m(p) - w_m(p)$ were nonzero, its holomorphic coefficient would have a positive lower bound on a smaller neighborhood. The finite weighted integral in Equation (2.12) would then contradict Equation (2.6). Removing the analytic null set $V \setminus U$ does not change this divergence. This also applies when p lies on B^- , since both sections are now holomorphic at p . We conclude

$$w_m(p) = s_m(p) \neq 0. \quad (2.15)$$

The bounded factor denoted $O_m(1)$ is used only to test integrability for a fixed m ; it never enters Equation (2.13).

It remains to convert that quadratic estimate to the original gauge. Hölder's inequality gives

$$Q_m(w_m)^m \leq C' \varepsilon_m \left(\int_U e^{(G-a-mb)/(m-1)} \right)^{m-1}. \quad (2.16)$$

There is no unbounded factor hidden in the second integral. Put $\eta = d/(m-1) \in (0, 1)$. Direct expansion yields

$$\frac{G-a-mb}{m-1} = (1-\eta)(\tau_m - a/m - b) + \eta(\max\{\tau_m - a/m, \phi\} - b). \quad (2.17)$$

Both exponentials on the right are canonical densities. The first has integral one; the second has integral at most $1 + I$, where

$$I = \int_V e^{\phi-b} < \infty.$$

Finiteness follows from local upper bounds for ϕ and local integrability of e^{-b} on a fixed finite coordinate cover. Applying Hölder to Equation (2.17) gives

$$\left(\int_U e^{(G-a-mb)/(m-1)} \right)^{m-1} \leq (1+I)^{\eta(m-1)} = (1+I)^d.$$

All integrals are unchanged by the omitted analytic null set. Together with Equations (2.5) and (2.16), this proves

$$Q_m(w_m)^m \leq C' \varepsilon_m (1+I)^d \rightarrow 0.$$

For large m , therefore, $0 < Q_m(w_m) < 1$. Multiply w_m by $Q_m(w_m)^{-m/2} > 1$. The new section has gauge one and, by Equation (2.15), strictly larger absolute evaluation than s_m . This contradicts the choice of s_m , regardless of the rate at which $s_m(p)$ or ε_m tends to zero. Hence ϕ has no positive Lelong number at any point. This proves Theorem 1.1.

Corollary 2.4. *Under the hypotheses of Theorem 1.1, let ϕ be a minimal weight on the actual rational line bundle N . Then*

$$\mathcal{I}(t\phi) = \mathcal{O}_V \quad \text{for every real } t > 0,$$

where the ideal is defined by local integrability of $|f|^2 e^{-t\phi}$. The conclusion is independent of the choice of minimal metric and of the Cartier multiple used to normalize its weights.

Proof. Skoda's integrability theorem gives local integrability of $e^{-t\phi}$ for every fixed $t > 0$ when the Lelong numbers vanish [21, Proposition 7.1]; for this formulation for arbitrary plurisubharmonic weights, see [5, Property 1.4(8)] and [4, Lemma 5.6(a)]. Every holomorphic germ is bounded on a smaller neighborhood, so the multiplier ideal is the structure sheaf. Two metrics with minimal singularities have weights differing by a bounded function, directly from their defining comparison. Dividing the weight of a positive Cartier multiple by that multiple preserves the stated normalization. These observations prove the independence assertions. $\square$

3 An interior injectivity theorem

We prove Theorem 1.2 from its two endpoint metrics, using the top-degree L^2 estimate in Lemma 2.1. The proof does not use Theorem 1.1 or its extremal-section construction. The output is the natural map on ordinary coherent cohomology. The proof therefore needs both a weighted estimate on the complement of the crossing divisor and a comparison that retains the original ordinary class. The fixed-cover construction below supplies this comparison together with a uniform bound for primitives.

For a class in the kernel, we will construct weighted harmonic representatives whose images have norms tending to zero. The endpoint metrics make multiplication a contraction and supply the curvature comparison for this estimate. Local solutions of the $\bar{\partial}$ equation leave holomorphic differences on overlaps; controlling a splitting of these differences on one fixed cover gives uniformly bounded global primitives. The extended differences form a cocycle representing the ordinary cohomology class of the original form. Finally, comparison in one fixed Hilbert space lets the small images force the original ordinary class to vanish. Classical injectivity for roots of divisors, including an additional reduced boundary, is proved in [7, Theorem 5.1]. The local and uniform-primitive methods have precedents in [19, Sections 5.2–5.3]; the precise comparison needed for the present line bundles is proved here.

3.1 Endpoint regularization and the comparison weights

Fix a Kähler form ω on V , and put $n = \dim V$. The zero-dimensional assertion is trivial, so assume $n > 0$. We use the following consequence of regularization and exponential integrability.

Lemma 3.1. *Let F be a rational line bundle on a smooth projective complex variety V equipped with a Kähler form ω , and suppose that F has a semipositive singular metric ψ having zero Lelong numbers everywhere. There are smooth weights ψ_k on F and numbers $\epsilon_k \downarrow 0$ such that*

$$i\partial\bar{\partial}\psi_k \geq -\epsilon_k\omega.$$

In fixed local frames these weights are uniformly bounded above on relatively compact coordinate neighborhoods, and, for every fixed $b > 0$, their exponentials $e^{-b\psi_k}$ have uniformly bounded local L^1 norms.

Proof. Choose a smooth reference metric on F . Apply the attenuation form of Demailly's regularization theorem to the specified current $T = i\partial\bar{\partial}\psi$, using the global difference between ψ and the reference weight as its potential [3, Main Theorem 1.1]. Fix a positive attenuation level c . Adding the reference weight back to the approximating potentials gives weights ψ_k on the same rational bundle, with $\psi_k \downarrow \psi$. They are smooth outside the Lelong level set $E_c(T) = \{x : \nu(T, x) \geq c\}$, which is empty. Their curvature lower bounds have the form $-\min(\lambda_k, c)u - \delta_k\omega$, where u is one fixed smooth nonnegative form, $\delta_k \rightarrow 0$, and the

continuous functions λ_k decrease to the Lelong-number function. That function is zero here. Dini's theorem on the compact manifold gives $\sup_V \lambda_k \rightarrow 0$, and $u \leq C\omega$ gives the stated loss $\epsilon_k \rightarrow 0$. Passing to a subsequence makes the losses decrease. Work first on a common integral tensor power of F and divide all weights by that integer afterwards.

On each fixed coordinate neighborhood the decreasing approximants satisfy $\psi \leq \psi_k \leq \psi_1$. The upper bound is smooth on a slightly larger neighborhood, and hence is bounded on the chosen compact subset. For every fixed $b > 0$,

$$e^{-b\psi_k} \leq e^{-b\psi}.$$

The right side is locally integrable by the zero-Lelong case of Skoda's integrability criterion [21]; see [5, Property 1.4(8)] for the formulation for arbitrary plurisubharmonic weights. This proves the uniform fixed-exponent bounds directly. $\square$

Apply the lemma simultaneously to $L - C_0$ and $L - C_2$, obtaining weights $\psi_{0,k}, \psi_{2,k}$ with a common sequence $0 < \epsilon_k \leq 1$ tending to zero. Let

$$D = \text{Supp}(C_0 + C_2)_{\text{red}}, \quad U = V \setminus D.$$

For a rational divisor C , write $[C]$ for its divisor weight. The endpoint weights $\psi_{i,k}$ live on the rational bundles $L - C_i$. Adding $[C_i]$ gives weights on L . We first raise the weight at the first endpoint above that at the second, so that the natural inclusion of the eventual integral bundles will be a contraction. The symbols $\Phi_{i,k}$ below are weights on L , whereas $h_{i,k}$ are weights on $L_i = L - [C_i]$. On U set

$$\psi'_{0,k} = \log(e^{\psi_{0,k}} + e^{\psi_{2,k} + [C_2 - C_0]}), \quad (3.1)$$

$$\Phi_{0,k} = \psi'_{0,k} + [C_0], \quad \Phi_{2,k} = \psi_{2,k} + [C_2], \quad \Phi_{1,k} = (1 - \lambda)\Phi_{0,k} + \lambda\Phi_{2,k}, \quad (3.2)$$

$$h_{i,k} = \Phi_{i,k} - [[C_i]], \quad i = 0, 1. \quad (3.3)$$

The two summands in Equation (3.1) are weights on the same rational line bundle $L - C_0$, so the expression is intrinsically well-defined. The logarithm of a sum of exponentials preserves the common curvature lower bound: locally add a potential for $\epsilon_k\omega$ to both summands and use the psh log-sum inequality. Moreover $\psi'_{0,k} \geq \psi_{0,k}$. Effectivity of $C_2 - C_0$ gives a uniform local upper bound for its divisor weight, so $\psi'_{0,k}$ retains both the upper bounds and all the fixed-exponent integrability bounds of Lemma 3.1.

The weights $h_{i,k}$ are smooth metrics on $L_i|_U$. They have uniform local upper bounds in frames extending across D , and

$$\sup_k \int_{B \cap U} e^{-h_{i,k}} d\lambda_B < \infty \quad (i = 0, 1) \quad (3.4)$$

on every relatively compact coordinate neighborhood B , where $d\lambda_B$ is coordinate volume. To verify the last assertion, write

$$h_{0,k} = \psi'_{0,k} + [C_0 - [C_0]], \quad h_{1,k} = (1 - \lambda)\psi'_{0,k} + \lambda\psi_{2,k} + [C_1 - [C_1]].$$

Every coefficient of either fractional divisor lies in $[0, 1)$. Choose one Hölder exponent $a > 1$ so close to one that its product with each of these finitely many coefficients is still less than one. The exponential of the fractional divisor weight is in L^a by the crossing-coordinate integral. The conjugate Hölder exponent for the remaining weights is allowed by the fixed-exponent bounds; a second Hölder inequality handles the two summands in $h_{1,k}$. This proves Equation (3.4) uniformly in k .

Let $E = [C_1] - [C_0] \geq 0$, and let $s : L_1 \rightarrow L_0$ be multiplication by the canonical section of E . Since $\Phi_{2,k} \leq \Phi_{0,k}$, we have $\Phi_{1,k} \leq \Phi_{0,k}$ and

$$|s|^2 e^{h_{1,k} - h_{0,k}} = e^{\Phi_{1,k} - \Phi_{0,k}} \leq 1. \quad (3.5)$$

Thus multiplication is a contraction on all form degrees for any one fixed background metric. On U the curvatures are $\theta_{i,k} = i\partial\bar{\partial}\Phi_{i,k}$; divisor weights contribute no curvature there.

3.2 A complete metric with bounded local potentials

The singular set must be removed to use smooth weighted harmonic forms. We choose the complete metric carefully so that local solvability still has a uniform comparison with the given weights. The bounded-potential construction follows the complete-metric method of [10, Lemma 3.1]; we include the calculation because both bounded potentials and completeness are needed in the uniform primitive estimate.

Lemma 3.2. *There is a complete Kähler metric $\omega_c \geq \omega$ on U such that, on sufficiently small coordinate neighborhoods in V , it has potentials bounded even on approach to D .*

Proof. If D is empty, take $\omega_c = \omega$. Otherwise choose a smooth metric on $\mathcal{O}_V(D)$ and scale it so that $t = \log |s_D|_{\text{sm}}^2 < -e^2$ on U . Set $f_0(t) = 1/\log(-t)$. For $x = -t$,

$$f'_0(t) = \frac{1}{x(\log x)^2}, \quad f''_0(t) = \frac{1}{x^2} \left(\frac{1}{(\log x)^2} + \frac{2}{(\log x)^3} \right).$$

In particular f'_0 is bounded and $f''_0(t) \geq (t \log(-t))^{-2}$. Off D the form $i\partial\bar{\partial}t$ is the negative of a fixed smooth curvature form, so it is bounded by a multiple of ω . For a sufficiently large fixed M the form

$$\omega_c = M\omega + i\partial\bar{\partial}f_0(t)$$

is positive and dominates both ω and $i\partial t \wedge \bar{\partial}t/(t \log(-t))^2$.

A path leaving every compact subset of U approaches D , because V is compact; along such an approach $t \rightarrow -\infty$. The displayed radial term bounds its length below by a positive constant times the total variation of $\log \log(-t)$, which diverges. Hence the metric is complete, including at crossings of components of D . Finally, if $\omega = i\partial\bar{\partial}\rho$ on a small coordinate neighborhood, then $M\rho + f_0(t)$ is a local potential for ω_c . The smooth function ρ is bounded on a smaller relatively compact neighborhood, and $f_0(t)$ is bounded and tends to zero at D . $\square$

Fix this metric for the rest of the section. When $D \neq \emptyset$, the function $r = \log \log(-t)$ is a smooth proper exhaustion of U with bounded gradient in ω_c . If ζ is a smooth function equal to one on $(-\infty, 1]$ and zero on $[2, \infty)$, then $\zeta(r/R)$, as $R \rightarrow \infty$, gives compactly supported smooth cutoffs tending to one and with gradient $O(R^{-1})$. When D is empty, the constant cutoff one suffices.

The curvature comparison following Equation (3.2) becomes

$$\theta_{1,k} + \epsilon_k \omega_c \geq (1 - \lambda)(\theta_{0,k} + \epsilon_k \omega_c) \geq 0, \quad (3.6)$$

since $\theta_{2,k} + \epsilon_k \omega_c \geq 0$ as well.

3.3 A fixed-cover primitive estimate and the ordinary class map

Fix a holomorphic line bundle F on V . The local L^2 theorem gives primitives on coordinate neighborhoods, but a global primitive requires their holomorphic differences on overlaps to split. We use one finite cover and its fixed spaces of holomorphic sections to make that splitting estimate uniform in the weights. The extended differences also define a class in the ordinary group $H^1(V, K_V \otimes F)$. Thus the class map below has ordinary coherent cohomology as its target, and the primitive estimate shows that its kernel consists of actual weighted exact forms. This local-solutions and holomorphic-cocycle comparison follows the method of [10, Section 3, Claim 1 and Lemma 3.2]. For a smooth metric h on $F|_U$, write $L_h^2(n, q)$ for the Hilbert space of F -valued (n, q) forms using h and ω_c . The Dolbeault operator is its maximal distributional realization, and $Z_h^1 = \ker(\bar{\partial} : L_h^2(n, 1) \rightarrow L_h^2(n, 2))$.

Proposition 3.3. *Let F be a fixed holomorphic line bundle on V , and let h_k be smooth metric weights on $F|_U$. Suppose that, for a fixed constant $A_0 \geq 0$,*

$$i\partial\bar{\partial}h_k \geq -A_0\omega_c,$$

and that in local frames on V the weights are uniformly bounded above and the functions e^{-h_k} have uniformly bounded local integrals on punctured coordinate neighborhoods. Then:

- (i) *There is a continuous linear map*

$$\kappa_k : Z_{h_k}^1 \longrightarrow H^1(V, K_V \otimes F)$$

agreeing with the ordinary Dolbeault class on smooth closed forms on V . It annihilates the closure in $L_{h_k}^2(n, 1)$ of the image of $\bar{\partial} : L_{h_k}^2(n, 0) \rightarrow L_{h_k}^2(n, 1)$.

- (ii) *There is a constant C independent of k such that every $f \in Z_{h_k}^1$ with $\kappa_k(f) = 0$ has a global primitive on U satisfying*

$$\bar{\partial}v = f, \quad \|v\|_{h_k} \leq C\|f\|_{h_k}.$$

No continuous linear choice of these global primitives is asserted.

Proof. Choose a finite collection of coordinate and frame balls

$$V_j \Subset U_j \Subset \hat{U}_j$$

such that the V_j still cover V and ω_c has a bounded local potential ρ_j on $\hat{U}_j \cap U$. These sets and potentials are independent of k .

Local solutions. On $U_j \cap U$, replace the line weight by $h_k + (A_0 + 1)\rho_j$. Its curvature dominates ω_c , and the bounded potential compares its top-form norms with those of h_k by fixed constants. Equip $U_j \cap U$ with the complete metric $\omega_c + \omega_{\text{ball}}$, where ω_{ball} is a complete metric for the coordinate ball. This sum is complete at both the deleted divisor and the ball boundary. An $(n, 1)$ datum square integrable for ω_c remains so for the larger background metric: its density uses the inverse background matrix on its one remaining covector. By Lemma 2.1, the inverse-curvature density for the modified weight is bounded by the original ω_c datum density. The top-form solution density is independent of the auxiliary complete metric. Consequently there is a local solution operator $T_{j,k}$ with

$$\bar{\partial}T_{j,k}f = f, \quad \|T_{j,k}f\|_{h_k, U_j \cap U} \leq C_j\|f\|_{h_k, U_j \cap U}, \quad (3.7)$$

where C_j is independent of k . For each fixed k , choose the solution of smallest norm in the modified metric. Orthogonal projection onto the complement of the holomorphic

kernel makes this a linear continuous choice. The affine solution set is closed because the distributional operator is closed.

Holomorphic cocycles. Given $f \in Z_{h_k}^1$, put $w_j = T_{j,k}(f|_{U_j \cap U})$. On $(U_i \cap U_j) \cap U$, the difference $w_j - w_i$ is holomorphic. Uniform upper bounds for h_k turn Equation (3.7) into ordinary local L^2 bounds in frames on V . Holomorphic L^2 removal therefore extends the differences uniquely across D . One can see the removal locally by Laurent expansion in the crossing coordinates and Fubini: a nonzero negative power is not square integrable. The extended differences $c_{ij} = w_j - w_i$, sections of $K_V \otimes F$ on $U_i \cap U_j$, form a holomorphic Čech cocycle.

For later use, this family of cocycles is bounded in every fixed compact-convergence seminorm whenever $\|f\|_{h_k}$ is bounded, uniformly in k . Indeed, for any compact set in an overlap, choose a slightly larger compact neighborhood still in that overlap. The ordinary L^2 bound there and the holomorphic mean-value inequality bound the supremum on the smaller compact. Each such compact has positive separation from the ball boundaries. Its constant may depend on that compact, but not on k . No bound on the supremum over an entire open overlap is required.

The ordinary class. Let $\mathcal{Z}^1$ be the Fréchet space of holomorphic cocycles on this finite cover, with compact-convergence topology. The cocycle equations make it a closed subspace of the finite product of overlap section spaces. Choose a smooth partition of unity (η_i) with $\text{Supp } \eta_i \subseteq U_i$. For a cocycle c put $b_j = \sum_i \eta_i c_{ij}$ on U_j , extending each summand by zero outside its overlap. Then $b_j - b_i = c_{ij}$, so the forms $\bar{\partial} b_j$ agree and define a global smooth form β_c . This defines the continuous linear map

$$\kappa : \mathcal{Z}^1 \longrightarrow H^1(V, K_V \otimes F), \quad c \longmapsto [\beta_c].$$

Continuity follows from the compact support of the partition and Cauchy estimates for derivatives on slightly larger compact sets; ordinary cohomology is finite-dimensional and Hausdorff on the compact manifold. Define $\kappa_k(f) = \kappa((c_{ij}))$. Changing the local primitives changes the cocycle by a holomorphic coboundary: the differences of two primitives extend by the same ordinary L^2 argument. The local solution estimates and the compact-seminorm estimates just proved make κ_k continuous and linear.

If f is smooth on V , choose smooth local primitives on the larger balls. On the smaller balls they have finite weighted norms, because their coefficients are bounded and the local integrals of e^{-h_k} are finite. Comparison with these primitives shows that κ_k agrees with the ordinary Dolbeault class. If instead $f = \bar{\partial} v$ for a global weighted L^2 primitive on U , then $w_j - v$ is holomorphic on $U_j \cap U$ and ordinary locally L^2 . Its extension gives a holomorphic splitting of c_{ij} , so $\kappa_k(f) = 0$. Continuity therefore annihilates the closure of the actual exact forms, as in (i).

A uniformly bounded splitting. Let $\mathcal{C}^0 = \prod_j H^0(U_j, K_V \otimes F)$, again with compact-convergence topology, and let $\mathcal{B}^1$ be the image of its coboundary map δ in $\mathcal{Z}^1$. We need the equality

$$\mathcal{B}^1 = \ker \kappa. \tag{3.8}$$

The forward inclusion is immediate. For the reverse inclusion, construct smooth local cochains from a partition of unity for a cocycle in $\ker \kappa$. Their common $\bar{\partial}$ is a global smooth form with zero ordinary Dolbeault class. Subtract a global smooth primitive from every cochain. The resulting cochains are holomorphic and split the original cocycle. This proves Equation (3.8) without any assumption that the coordinate cover is a Leray cover.

It follows that $\mathcal{B}^1$ is a closed Fréchet subspace. The continuous surjection $\delta : \mathcal{C}^0 \rightarrow \mathcal{B}^1$ is open. Prescribe only the single continuous seminorm

$$q((a_j)) = \max_j \sup_{\bar{V}_j} |a_j|,$$

where the norms are those of fixed smooth frames or a fixed smooth bundle metric. Openness says that the image of $\{q < 1\}$ contains a zero neighborhood of $\mathcal{B}^1$ described by finitely many compact seminorms. The cocycles arising from all k and all data of norm at most one are bounded in those seminorms. A single rescaling therefore gives, for every such cocycle in $\mathcal{B}^1$, a splitting (a_j) with $q((a_j)) \leq C_{\text{split}}$ for a fixed C_{split} . This uses no assertion that a bounded subset has a lift bounded in every Fréchet seminorm.

When $\kappa_k(f) = 0$, choose this splitting and set $v = w_j - a_j$ on $U_j \cap U$. The expressions agree on the full overlaps, and hence define a global primitive on U . Estimate its norm on the smaller V_j , which cover V . The local primitives are controlled by Equation (3.7); the corrections are controlled by their supremum on $\overline{V_j}$ and the uniform integral of e^{-h_k} there. Summing over the finite cover proves a uniform bound for unit-norm data. Scaling proves (ii). $\square$

Both sequences $h_{0,k}$ and $h_{1,k}$ satisfy this proposition, by Equations (3.4) and (3.6). For each weight, parts (i) and (ii) give

$$\ker \kappa_k = \text{im } \bar{\partial} = \overline{\text{im } \bar{\partial}} \quad \text{inside } Z_{h_k}^1.$$

Indeed, part (i) kills the closure of the exact forms, while part (ii) provides a primitive for every form in the kernel. We will keep track of the closures explicitly when comparing different weighted spaces. The proposition also applies to a fixed smooth line metric from V : its curvature is bounded below by $-A_0\omega_c$ for some fixed A_0 , and its local upper and integral bounds are automatic. This last observation will allow us to compare all approximation parameters in one Hilbert space.

3.4 Harmonic representatives and the Bochner comparison

Take a class in the kernel of the map in Theorem 1.2, and represent it by a smooth closed L_1 -valued $(n, 1)$ form γ on V . All norms in this subsection use ω_c ; a subscript i, k indicates the line weight $h_{i,k}$, and integrals of pointwise inner products use dV_{ω_c} . The norms $\|\gamma\|_{1,k}$ are bounded uniformly. Indeed, enlarging ω to ω_c decreases the top-form-valued $(0, 1)$ density, and the coefficients of γ are bounded on a finite cover, where Equation (3.4) applies.

In the maximal $L_{1,k}^2$ Dolbeault complex, the closure of the image of degree-zero forms lies in the closed kernel of the degree-one operator. Subtract its orthogonal projection from γ , obtaining u_k . Thus

$$\gamma - u_k \in \overline{\text{im } \bar{\partial}}, \quad \bar{\partial}u_k = 0, \quad \bar{\partial}_{1,k}^*u_k = 0, \quad \|u_k\|_{1,k} \leq \|\gamma\|_{1,k}. \quad (3.9)$$

The adjoint assertion follows from orthogonality to the actual range; it does not require that range to be closed. Local elliptic regularity for the smooth weighted Dolbeault Laplacian on U makes u_k smooth there.

Write $\Lambda = \Lambda_{\omega_c}$ and let $D'_{i,k}$ be the $(1, 0)$ Chern derivative. In bidegree $(n, 1)$ put

$$A_{i,k} = [\theta_{i,k} + \epsilon_k \omega_c, \Lambda].$$

These are pointwise nonnegative operators, and $A_{1,k} \geq (1 - \lambda)A_{0,k}$. The smooth compact-support Bochner–Kodaira identity is

$$\|\bar{\partial}v\|_{i,k}^2 + \|\bar{\partial}_{i,k}^*v\|_{i,k}^2 = \|D'_{i,k}v\|_{i,k}^2 + \int_U \langle [\theta_{i,k}, \Lambda]v, v \rangle_{i,k}. \quad (3.10)$$

Here $D'_{i,k}v = 0$ in top holomorphic degree, and $[\omega_c, \Lambda]$ is the identity in this bidegree. The same identity on complete manifolds, with the indicated domain interpretation and lower curvature bound, is discussed in [19, Proposition 2.4].

To justify its use without already knowing the extra terms are integrable, use the compactly supported smooth cutoffs constructed above, whose gradients tend uniformly to zero in ω_c . Apply Equation (3.10) to the cutoff multiples of u_k and add ϵ_k times their squared norms. The left-hand derivative terms tend to zero by Equation (3.9) and the gradient bound. The two terms retained on the right are nonnegative, so lower semicontinuity gives

$$\|D'_{1,k} u_k\|_{1,k}^2 + \int_U \langle A_{1,k} u_k, u_k \rangle_{1,k} \leq \epsilon_k \|u_k\|_{1,k}^2. \quad (3.11)$$

The commutators with the cutoffs contain only their derivatives; they do not introduce constants from derivatives of the varying line metrics.

The Kähler identity $D'^* = i[\Lambda, \bar{\partial}]$ shows that this formal operator commutes with holomorphic multiplication by s . Combining Equations (3.5) and (3.6) with Equation (3.11) therefore gives

$$\|D'_{0,k}(su_k)\|_{0,k}^2 + \int_U \langle A_{0,k} su_k, su_k \rangle_{0,k} \leq C_\lambda \epsilon_k \|u_k\|_{1,k}^2. \quad (3.12)$$

For the curvature term one first uses $A_{1,k} \geq (1 - \lambda)A_{0,k}$ and then the line-norm contraction. One may take $C_\lambda = (1 - \lambda)^{-1}$: this factor bounds both the derivative term and the curvature term in the source energy. The strict interior hypothesis keeps this comparison factor finite.

We also need the target $\bar{\partial}$ adjoint. The form su_k is smooth, closed, and square integrable. Apply the compact-support identity to cutoff multiples of this form. On its right-hand side use Equation (3.12), replace the curvature commutator by the larger nonnegative $A_{0,k}$, and let the gradient errors tend to zero. It follows that the formal adjoint is square integrable and

$$\|\bar{\partial}_{0,k}^*(su_k)\|_{0,k} \leq C\sqrt{\epsilon_k}. \quad (3.13)$$

This also verifies the Hilbert-adjoint domain condition. In detail, complete-metric cutoffs and local smooth approximation give density in the maximal graph norm, so integration by parts against compactly supported forms extends to that domain. The formal adjoint, now known to be L^2 , is its Hilbert adjoint. Thus Equation (3.13) has not assumed the existence of a global primitive or an adjoint-domain statement in advance.

We have shown that the image of the harmonic representative has a small adjoint norm. The fixed-cover proposition supplies a uniformly bounded primitive for this image; pairing these two statements will make its entire target norm tend to zero.

3.5 From a small image to vanishing of the ordinary class

Multiplication by s is bounded in degrees zero and one and commutes with the distributional Dolbeault operator. Hence Equation (3.9) implies that $s\gamma - su_k$ is in the target closure of the actual exact forms. Apply the target class map in Proposition 3.3. It annihilates that closure and agrees with ordinary cohomology on $s\gamma$. The latter class is zero by the choice of γ . Consequently $\kappa_{0,k}(su_k) = 0$.

The norms of su_k are uniformly bounded by Equations (3.5) and (3.9). Part (ii) of Proposition 3.3 gives global forms v_k on U with

$$\bar{\partial}v_k = su_k, \quad \sup_k \|v_k\|_{0,k} < \infty.$$

The adjoint-domain conclusion above now justifies

$$\|su_k\|_{0,k}^2 = \langle \bar{\partial}v_k, su_k \rangle_{0,k} = \langle v_k, \bar{\partial}_{0,k}^*(su_k) \rangle_{0,k} \longrightarrow 0. \quad (3.14)$$

To retain the original class while passing to a limit, fix a smooth metric on L_1 over all of V , with local weight h_* and use the same complete background ω_c on U . Uniform upper bounds in a finite framed cover give the global metric comparison

$$h_{1,k} \leq h_* + C_*, \quad e^{-h_*} \leq e^{C_*} e^{-h_{1,k}}. \quad (3.15)$$

It gives one bounded inclusion from each varying weighted space into the fixed space, in both degrees zero and one. In particular the u_k have uniformly bounded fixed norms.

On any compact subset $K \subset U$ the multiplier s is nowhere zero, and the target weights $h_{0,k}$ have uniform upper bounds. Thus Equation (3.14) implies that $u_k \rightarrow 0$ in the fixed L^2 norm on K . This local strong convergence and the global bound imply global weak convergence to zero. Indeed, approximate any fixed L^2 test form by one supported in a compact subset of U ; its pairing tends to zero by local convergence, and the norm of the remaining tail controls its pairing uniformly in k . Concentration of norm near D does not affect this weak convergence.

For each k , choose actual exact forms $\bar{\partial}z_{k,l}$ converging to $\gamma - u_k$ in Equation (3.9). By Equation (3.15), their primitives belong to the fixed degree-zero space and their derivatives converge in the fixed degree-one norm. Therefore $\gamma - u_k$ belongs to the fixed closure of actual exact forms. Apply Proposition 3.3 once more, now to the fixed smooth metric with weight h_* . Its continuous class map κ_* annihilates that fixed closure, so

$$\kappa_*(u_k) = \kappa_*(\gamma) = [\gamma] \quad \text{for every } k.$$

The fixed closed-form space is a closed Hilbert subspace, and a continuous linear map from it to finite-dimensional ordinary cohomology is weakly continuous. Since $u_k \rightarrow 0$, the last displayed identity forces $[\gamma] = 0$.

This proves analytic Dolbeault injectivity. Dolbeault theory and GAGA identify it with the natural map in algebraic coherent cohomology, proving Theorem 1.2.

4 Abundance for fourfolds with nonzero Euler characteristic

The metric theorem gives abundance for klt fourfolds when the Euler characteristic is nonzero. A published nonvanishing criterion uses this Euler hypothesis to produce the first section; a second criterion then gives semiample. We state both inputs in terms of the actual adjoint bundle on a resolution, so that the same zero-Lelong metric supplies their analytic hypotheses.

Corollary 4.1. *Let (X, Δ) be a projective complex klt fourfold with effective rational boundary and nef adjoint $D = K_X + \Delta$. If $\chi(X, \mathcal{O}_X) \neq 0$, then D is semiample.*

This proof combines Theorem 1.1 with published nonvanishing and semiample criteria. It isolates the use of $\chi(X, \mathcal{O}_X) \neq 0$: that condition enters the nonvanishing criterion, whereas the metric theorem has no Euler-characteristic hypothesis. Related work on multiplier-ideal approximations and supercanonical currents studies this analytic setting [16, Theorems A–B].

The two algebraic inputs

For a $\mathbb{Q}$ -Cartier divisor P on a normal projective variety, the metric condition in [17, Definition 2.12] requires a positive integer r , with rP Cartier, and a resolution $f : Z \rightarrow X$ carrying a singular metric on $f^*\mathcal{O}_X(rP)$ whose curvature has the form

$$T = T_0 + \sum_j a_j [E_j], \quad T_0 \geq 0, \quad \nu(T_0, z) = 0 \quad (z \in Z), \quad a_j \in \mathbb{Q}_{\geq 0}. \quad (4.1)$$

The divisor sum is finite. This is called a metric with *generalized algebraic singularities*. The sum may be zero; in particular, every semipositive metric with zero Lelong numbers everywhere meets this definition.

Theorem 4.2 (Lazić–Petersen, fourfold nonvanishing). *Let (X, Δ) be a normal projective complex klt pair of dimension four, with effective rational boundary, and put $D = K_X + \Delta$. Suppose D is pseudoeffective and N is a nef $\mathbb{Q}$ -Cartier divisor. If both D and $D + N$ satisfy the metric condition (4.1) and $\chi(X, \mathcal{O}_X) \neq 0$, then some rational $t_0 > 0$ satisfies*

$$\kappa(X, D + tN) \geq 0 \quad \text{for every rational } t \in [0, t_0].$$

This is [17, Corollary D], the unconditional four-dimensional consequence of their Theorem C(i). Its interval includes $t = 0$. We will take $N = 0$, so both metric assumptions concern D .

Theorem 4.3 (Gongyo–Matsumura, fourfold semiample). *Let (X, Δ) be a normal projective complex klt pair of dimension four, with effective rational boundary, and set $D = K_X + \Delta$. Suppose that for a projective birational morphism $f : Z \rightarrow X$ with Z smooth and some positive integer r with rD Cartier, the bundle $f^*\mathcal{O}_X(rD)$ carries a semipositive singular Hermitian metric with zero Lelong numbers at every point of Z . If $\kappa(X, D) \geq 0$, then D is semiample.*

This is the form of [11, Corollary 5.3] used here. Positivity of the Cartier multiple is also explicit in their Theorem 5.1, from which that corollary follows. Neither of these fourfold results requires X to be $\mathbb{Q}$ -factorial or non-uniruled. Their dimension-three inputs are already incorporated in the stated corollaries.

Application to the minimal metric

Proof of Corollary 4.1. Choose a projective log resolution $\pi : Y \rightarrow X$, a positive integer r such that rD is Cartier, and a metric $h_{\min}$ with minimal singularities on $L = \pi^*D$. Nefness makes D pseudoeffective, so such a metric exists. By Theorem 1.1,

$$\nu(h_{\min}, y) = 0 \quad (y \in Y).$$

The tensor power $h_{\min}^{\otimes r}$ is a metric on the actual line bundle $\pi^*\mathcal{O}_X(rD)$. Its local weights are $r\phi$, so its curvature is semipositive and its Lelong numbers are $r\nu(\phi, y) = 0$.

Consequently, (4.1) holds with T_0 equal to this curvature current and with zero divisor sum. Apply Theorem 4.2 with $N = 0$ and $t = 0$ to obtain

$$\kappa(X, D) \geq 0.$$

This is nonvanishing for D itself: a positive Cartier multiple of D has a nonzero global section. It is therefore the Kodaira dimension hypothesis of Theorem 4.3. The same resolution and the same metric $h_{\min}^{\otimes r}$ supply its remaining hypothesis, and that theorem proves that D is semiample. $\square$

The argument imposes no restriction on the numerical dimension of D . The Euler-characteristic hypothesis is used only to obtain the first section; the interior-injectivity theorem of this article is not an input to this particular application.

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